INCH-POUND
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May 26,2020
SUPERSEDING
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August 11, 1992

COMMERCIAL ITEM DESCRIPTION

MAT, SLEEPING, SELF-INFLATING

The General Services Administration has authorized the use of this Commercial Item Description, for all federal agencies.

1. SCOPE. This commercial item description covers a self-inflating sleeping mat to insulate from the cold and moisture of bare ground when placed under a sleeping bag, and also to cushion soldiers from obtrusive objects when lying on the ground.

2. CLASSIFICATION.

2.1 Classes.

Class 1	Camouflage Green 483
Class 2	Foliage Green 504

3. SALIENT CHARACTERISTICS.

3.1 Description. The self-inflating sleeping mat shall be constructed of polyurethane coated nylon fabric fully bonded to a solid, continuous piece of open-cell, polyurethane foam (without die-cut holes). The seams of the mat shall be heat or dielectric sealed. The mat shall be self-inflating when the plastic air valve is opened. Deflation of the mat shall be achieved by compressing the mat and forcing the air out through the plastic valve. The upper and lower surfaces of the mat shall be free of obstructions or protrusions of any kind. Each mat shall come with a repair kit consisting of four (4), 2-3/4 inch diameter, nylon patches with color matching the mat, one (1) tube containing 0.20 fluid ounces of cement, and a repair instruction sheet. Care and use instruction shall be furnished with each mat.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: DLA Troop Support, Clothing and Textiles Directorate, Attn: DLA Troop Support Standardization Team, 700 Robbins Street, Philadelphia, PA 19111.

AMSC N/A FSC 8465

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3.2 Label. A commercial type label containing the contractor's name, contract number, National Stock Number and care instructions is required. In addition, this information may be applied by any means that provides the required degree of legibility and durability. The label/markings shall be permanently affixed to the sleeping mat. The color of the label shall approximate the color of the base material of the sleeping mat. The markings shall be Black, indelible ink, and shall not show offsetting, smearing or bleeding.

3.2.1 Bar Code Label/Tag. Each item shall be individually bar-coded with a Type VII, Class 17 label/tag of MIL-DTL-32075. The bar coding element shall be a 13 digit NSN. The label/tag shall be located so that it is completely visible on the item when it is folded and/or packaged as specified and shall cause no damage to the item.

3.3 Basic materials and components.

3.3.1 Coated cloth/foam laminated fabric. The mat shall be constructed of polyurethane coated, 200 Denier, oxford weave, nylon (Type 6/6) fabric weighing approximately 5.5 oz/sq. yd. For Class 1, the color shall match color Camouflage Green 483; for Class 2, the color shall match color Foliage Green 504 when tested in accordance with 5.5.3. The mat shall conform to spectral reflectance requirements of Table I when tested in accordance with 5.5.4. and infrared spectral reflectance requirements of Table II. The coated cloth shall be fully bonded to a solid, continuous piece of open-cell, polyurethane foam (without die-cut holes).

3.3.2 Valve. The air valve shall be made using acetal or nylon plastic materials. The valve shall be cylindrical in shape having a maximum, widest point diameter of 3/4-inch and a maximum length of 7/8-inch extending beyond the edge or surface of the sleeping pad. The valve shall be capable of being opened and securely closed in two revolutions of a non-removable, threaded cap with O-ring seal. For Class 1, the color of the valve shall match color Black or Camouflage Green 483; for Class 2, the color shall match color Foliage Green 504 when tested in accordance with 5.5.3. The mat shall conform to spectral reflectance requirements of Table I when tested in accordance with 5.5.4 and infrared spectral reflectance requirements of Table II.

TABLE I. Spectral Reflectance Requirements: Reflectance (percent) for Class 2, Foliage Green 504

Wavelength (nm)	Min	Max
600	8	18
620	8	18
640	8	20
660	10	26
680	10	26
700	12	28
720	16	30
740	16	30
760	18	32
780	18	34
800	20	36
820	22	38

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840	24	40
860	26	42

TABLE II. Infrared Spectral Reflectance Requirements for Foliage Green 504

Acetal or Nylon Plastic Hardware

Wavelength (nm)	Min	Max
600	8	18
620	8	18
640	8	18
660	10	26
680	10	26
700	12	28
720	20	36
740	26	40
760	30	52
780	32	56
800	32	60
820	34	60
840	36	60
860	38	60

3.4 Physical Requirements of End Item.

3.4.1 Air Retention. The sleeping mat shall not show evidence that air has leaked and visually inspect the test specimen for any apparent non-uniformity of thickness denoting evidence of separation of the foam and fabric surfaces or any openings at seams when tested in accordance with 5.5.1.

3.4.2 Dimensions. The uninflated sleeping mat shall be uniform in dimensions and have outside dimensions of 20-1/4 +/- 1/4 inches wide and 73-1/4 +/- 1/4 inches long measured from outside edges of seal. The fully-inflated mat shall be 1-inch to 1-1/4 inch thick with no load; The mat shall have a heat- or dielectric-sealed, lap seam of width of 1/4 +/- 1/16 inch; the inside edge of the sealed seam shall be not more than 3/8-inch to the edge of the foam. Examination shall be in accordance with 5.5.2.

3.4.3 Weight. The finished weight of the sleeping mat shall be less than 2.0 pounds when tested in accordance with 5.5.5.

3.4.4 Compressibility. The sleeping mat shall be capable of being rolled and compressed to a circumference of less than 16 inches when tested in accordance with 5.5.6.

3.4.5 Self-Inflation. The sleeping mat shall self-inflate within three (3) minutes when tested in accordance with 5.5.7.

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3.4.6 Recovery. The thickness of the sleeping mat shall be equal to or greater than 85 percent of the manufacturer's specified thickness when tested in accordance with 5.5.8.

3.4.7 Burst Pressure. The sleeping mat shall not show signs of delamination, foam rupture, or seam breakage when tested in accordance with 5.5.9.

3.4.8 Repair Kit. A repair kit shall be provided with each sleeping mat and shall consist of four (4), 2-3/4 inch diameter, nylon patches, one (1) tube containing 0.20 fluid ounces of cement, and a repair instruction sheet. The color of the nylon patches shall match the color of the sleeping mat. Care and use instruction shall also be furnished with each mat.

3.4.9 Toxicity. The finished sleeping mat shall not present a health hazard and shall show compatibility with prolonged, direct skin contact. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used. Toxicity requirements shall be tested in accordance with Modified Draize Procedure and 5.5.10.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery thereafter as may be otherwise provided for under the provisions of the contract.

5.3 Workmanship. The finished sleeping mat shall be thoroughly cleaned, and all loose thread, lint, and foreign matter removed.

5.4 Responsibility for Inspection. The supplier is responsible for assuring the product conforms to the requirements. Sleeping mat lots will be inspected to determine conformance with the Commercial Item Description and the successful offeror's technical proposal as negotiated. Acceptability of sleeping mat lots will be determined by the use of "Sampling Procedures and Tables for Inspection by Attributes", ASQ/ANSI Z1.4, published by the American Society for Quality Control. Sampling procedures will be designated by the Government at the time of award. The sample is one sleeping mat. The sleeping mat shall be examined for commercial type defects only.

5.5 Methods of Test.

5.5.1 Methods of testing, preconditioning and air retention. All testing shall be done in a standard condition environment defined by the ASTM D1776/D1776M, if not specifically defined by the

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individual test procedure. The test specimen shall be one, complete sleeping mat. All test specimens sampled for end item testing shall be preconditioned with initial inflation to a test pressure of 1.0 + 0.1 pounds per square inch (psi) for 24 hours; the test specimens shall then be deflated and tested for the required property. The air retention properties of the sleeping mat shall be determined using the preconditioned test specimens as follows: At the end of the inflation test period, inspect all the test specimens to determine that air has not leaked (the air pressure shall have not dropped more than 0.3 psi) and then visually inspect the test specimens for any evidence of openings at seams or any non-uniformity of thickness denoting separation of the foam and fabric surfaces. The lot shall be considered unacceptable if any test specimen fails; not less than five test specimens shall be examined.

5.5.2 Physical dimensions. The overall dimensions of the test specimen and the dimensions of the valve component, seal width and seal distance from foam shall be measured using standard rule to the nearest 1/16-inch. The actual thickness of the test specimen shall be the manufacturer's specified thickness, measured in inches.

5.5.3 Visual shade matching. The color and appearance of the cloth face and the valve of the test specimen shall match the standard sample when viewed using AATCC Evaluation Procedure 9, Option A, with sources stimulating artificial daylight D75 illuminant with a color temperature of 7500 + 200 K, with illumination of 100 + 20 foot-candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 + 200 K.

5.5.4 Spectral reflectance. Spectral reflectance data shall be obtained from 600 to 860 nanometers (nm) on a spectrophotometer relative to a barium sulfate standard, the preferred white reference standard. Other white reference materials may be used, provided they are calibrated to absolute white; e.g., magnesium oxide or vitrolite tiles. The spectral band width shall be less than 26 nm at 860 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either CIE Source A or CIE Source D65. The specimen shall be viewed at an angle no greater than 10 degree from normal, with the specular component included. Photometric accuracy of the spectrophotometer shall be within 1 percent and wavelength accuracy within 2 nm. The standard aperture size used in the color measurement device shall be 1.0 to 1.25 inches in diameter. For fabrics, test areas to be measured shall be at least 6 inches from the selvage. The specimen shall be measured as a single layer backed with layers of the same fabric and shade. For plastic components, test specimens shall be formed in flat, matte finish, molded blanks of the polymer used in the components and formed in a size to fit in the spectrophotometer specimen holder. Measurements shall be taken on a minimum of two different areas and the data averaged. When the measured reflectance values for the color at four or more wavelengths do not meet the limits, it shall be considered a test failure.

5.5.5 Weight. The finished weight of the test specimen shall be determined using a calibrated scale or balance and measured to the nearest 0.1 pound.

5.5.6 Compressibility. Inflate the test specimen with air to 1.0 + 0.1 psi, close the inflation valve and allow it to stand for 20 minutes. Then, open the valve and allow the test specimen to deflate

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and equalize for 10 minutes. With valve open, roll the test specimen by hand from the end opposite the valve, forcing air out through valve. Once rolled, close the valve and unroll the test specimen. With valve closed, roll the test specimen from the end opposite the valve a second time, forcing air to the upper portion of the mat (valve end). Once the test specimen has been rolled 50 percent of its length, open the valve with one hand while firmly compressing the half-rolled test specimen with the other hand. Continue tightly rolling the test specimen, forcing air out through the valve until the test specimen is completely rolled. Once rolled, close the valve and measure the circumference at the center of the rolled test specimen to the nearest 1/16-inch. Five test specimens shall be tested; failure of any test specimen to meet the requirement shall be cause for failure of the lot.

5.5.7 Self-Inflation. Roll the test specimen in accordance with procedure described in 3.4.4 and then wrap and secure it in its rolled form with a low-stretch cloth, 36-inches long by the full width of the test specimen. Open the inflation valve and condition the wrapped test specimen for 24 hours. At the end of the conditioning period, close the inflation valve and remove the cloth wrapping leaving the test specimen in its rolled form on a flat, level surface. Open the inflation valve and start a stopwatch timer. The timer shall be stopped at the moment that the test specimen has unrolled completely and the time recorded. Test specimens that do not unroll completely are considered as failing. Five test specimens shall be tested; failure of any test specimen to meet the requirement shall be cause for failure of the lot.

5.5.8 Recovery. The self-inflation of 3.4.5 shall be continued for each test specimen for 10 minutes total elapsed time. Close the inflation valve and measure the thickness of the test specimen using a 12-inch by 12-inch, calibrated Measure-Matic Thickness Gauge, or equivalent, under a pressure of 0.002 psi. The percent recovery shall be calculated by dividing the measured thickness by the manufacturer's specified thickness and expressed as a percentage recovery. Five test specimens shall be evaluated; failure of any test specimen to meet the requirement shall be cause for failure of the lot.

5.5.9 Burst Pressure. The burst pressure test shall be conducted using an air supply with pressure of at least 15 psi, an air regulator with a range of 2 to 250 psi (i.e., Type 70) and an air pressure gauge with a range of 0 to 15 psi, resolution of 0.1 psi, an accuracy of +/-0.25 percent of full scale and a capability of recording the peak pressure. The air pressure shall be applied to the sleeping mat using appropriate connectors, fittings and hoses. The test specimen shall be one complete sleeping mat. Open the inflation valve on the test specimen and allow the test specimen to fully self-inflate. Securely attach the air hose to the open inflation valve and slowly allow the air supply to flow into the test specimen until the pressure stabilizes at 3 psi. Once the pressure has stabilized, slowly increase the pressure in 0.5 psi increments allowing 15 seconds between increments. Continue increasing the air pressure until a pressure reading of 7 psi is reached and is maintained for 15 seconds; the test specimen considered passing. At each 15-second increment, examine the test specimen for any evidence of failure defined as evidence of delamination, foam rupture, or seam breakage/burst of the test specimen. Should the test specimen fail prior to meeting the passing criteria at the 7 psi air pressure, record the peak pressure and the mode of failure. Three test specimens shall be tested; failure of any test specimen to meet the requirement shall be cause for failure of the lot.

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5.5.10 Toxicity test. When required, an acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of these studies indicate the (item) is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 3.4.9). If the toxicity requirement can be demonstrated with historical use data, toxicity testing may not be required.

6 PACKAGING.

6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order.

7 NOTES.

7.1 Source of documents.

7.1.1 The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at http://www.arnet.gov/far.

7.1.2 DoD specifications may be obtained from https://quicksearch.dla.mil.

7.1.3 Copies of non-government documents may be obtained from: AATCC Standards are available online at www.aatcc.org or from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215. ASTM Standards may be obtained from: www.astm.org. Principles and Methods of Toxicology (fourth edition), may be obtained from: http://www.taylorandfrancis.co.uk/. ASQ/ANSI Z1.4 may be obtained from the American Society for Quality online at www.asq.org.

7.2 Source of standard samples. Standard samples may be obtained from https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx. Under "Special Instructions", provide color shade and roll number.

7.3 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Packaging requirements.

MILITARY INTERESTS

Custodians:

Army - GL
Navy - NU

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Review Activities:
Navy - MC

Preparing Activity:
DLA - CT
Project No. 8465-2019-004

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